Natural Language/Image Bird Search using Clip Embeddings

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The Problem

- Traditional image search relies on manually tagging images with relevant description, features and class.
- Doing this process by hand difficult and error prone and defeats the purpose of identifying an unknown bird species.

The Problem

- You may think to use a traditional classifier to automate the process of creating tags.

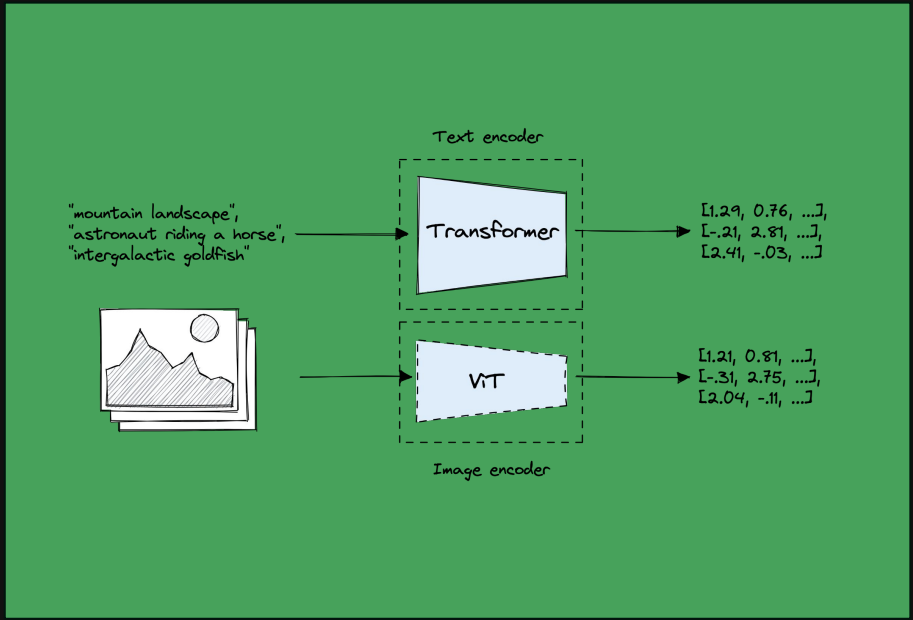
- However this approach still has several flaws.
  - Tag based search lacks relationship between words for example a sentence containing `red head` and `grey wing` could become confused and result in images of birds with a grey head and red wings.
  - Lacks the ability to search by image which is an essential feature for identifying birds.

The Problem

- A Clip based approach solves this issue by encoding the image and the query into an internal image embedding space.
- These embeddings (image or text) capture the entire image/prompt but can be simply compared using cosine similarity.
- This is the solution I will implement in this project

Model

- Training a Clip model from scratch would require to much compute and Data
- Instead I use the openai/clip-vit-base-patch32 model
- This Model has been pre-trained using > 400 million Image prompt pairs
- It consists of an text encoder (transformer) and image encoder (ViT) which are trained to have a shared embedding space.



Dataset

- CUB-200-2011 dataset
- Contains 11,788 images of birds from across 200 classes each image is tagged with a number of attributes (See example).
- The data is spread across a number of files which needed to be merged before it can be used for fine tuning.

```
attributes.txt
└── CUB_200_2011
    ├── attributes
    │   ├── certainties.txt
    │   ├── class_attribute_labels_continuous.txt
    │   └── image_attribute_labels.txt
    ├── bounding_boxes.txt
    ├── classes.txt
    ├── image_class_labels.txt
    ├── images
    │   ├── 001.Black_footed_Albatross
    │   ├── 002.Laysan_Albatross
    │   ├── 003.Sooty_Albatross
    │   ⋮          ⋱
    ├── images.txt
    ├── parts
    │   ├── part_click_locs.txt
    │   ├── part_locs.txt
    │   └── parts.txt
    ├── README
    └── train_test_split.txt
```

```
Sample Bird:
  Class: Yellow Warbler
  Attributes: leg color buff, forehead color grey, crown color yellow, primary color yellow, back color grey, breast color grey, size small (5 - 9 in)
  Prompt: Yellow Warbler with leg color buff, forehead color grey, crown color yellow, primary color yellow, back color grey, breast color grey, size small (5 - 9 in)
```

Fine Tuning

- To fine tune the model I would need some training examples which could be created from the data
- Prompt Templates
  - "a photo of a {class_name} with {attrs}"
  - "{class_name} with {attrs}"
  - "a {class_name}, has {attrs}"
  - "{class_name}, {attrs}"
- These prompts and images where then processed and used as examples for training
- The model was trained for 5 epochs at a learning rate of 5e-6

Application Pipeline

Training

The training was completed ahead of time using the jupyter notebook. It handles parsing the dataset and training and saving weights

Backend Server (Flask)

The backend loads the models weights from training, calculates the embedding and cosine similarity between all images and the query and return the top results to the frontend to be displayed

Front end

The front end is a simple web application which handles the user input, displays results and

Demo

Thank You!

Github:
https://github.com/Jordan-Brown15/natural-bird-search

Video:
https://youtu.be/CVYOU6NjTWg?si=OBSGZowQa0nBBdOV

References:

Radford et al., "Learning Transferable Visual Models From Natural Language Supervision", ICML 2021

Wah et al., "The Caltech-UCSD Birds-200-2011 Dataset", 2011